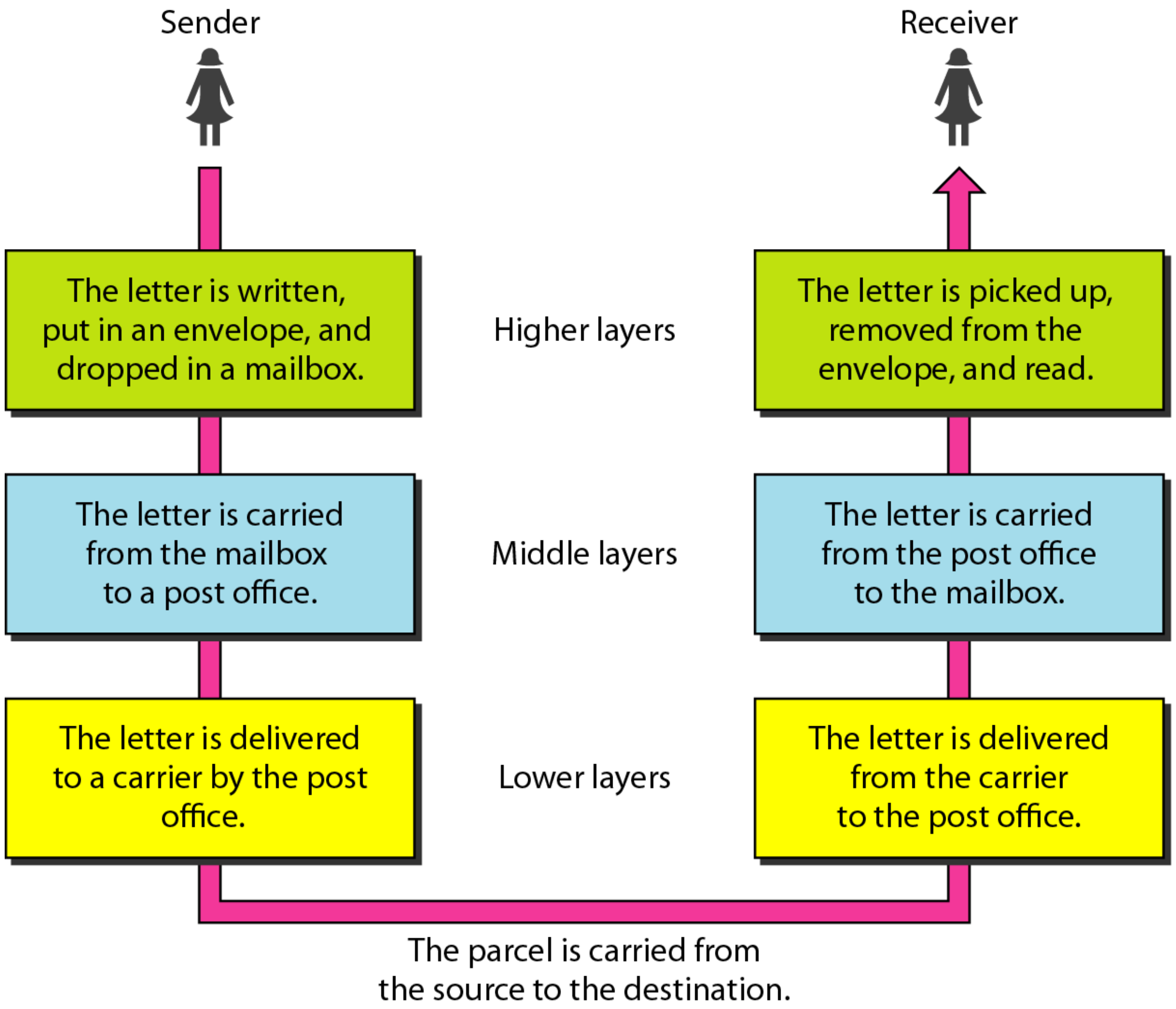
Computer Networks

Network Models

2-1 LAYERED TASKS

- We use the concept of layers in our daily life.
- As an example, let us consider two friends who communicate through postal mail.
- The process of sending a letter to a friend would be complex if there were no services available from the post office.

Figure 2.1 Tasks involved in sending a letter



2-2 THE OSI MODEL

- Established in 1947, **the International Standards Organization** (ISO) is a multinational body dedicated to worldwide agreement on international standards.
- An ISO standard that covers all aspects of network communications is the **Open Systems Interconnection** (OSI) model.
- It was first introduced in the late 1970s.

Note

ISO is the organization.
OSI is the model.

Figure 2.2 *Seven layers of the OSI model*

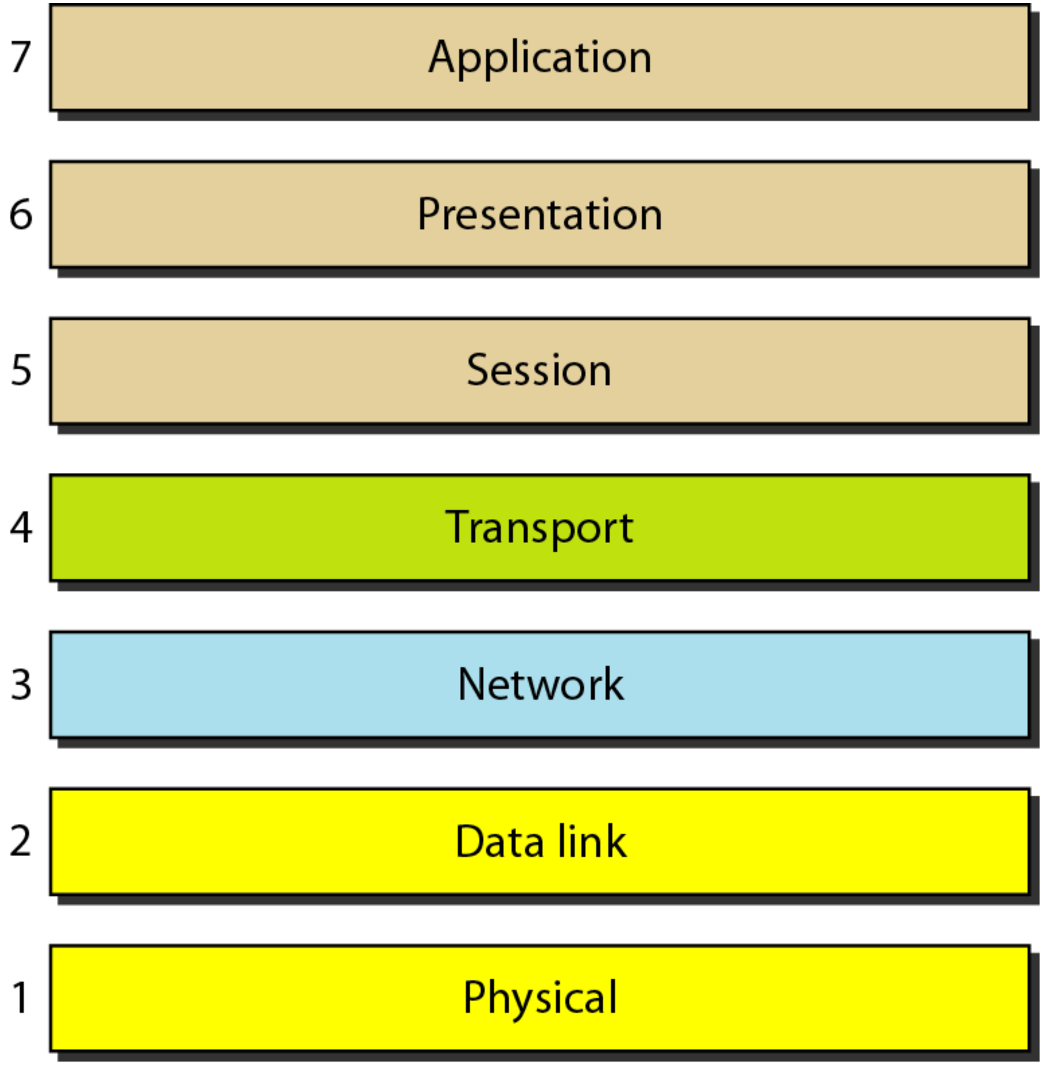


Figure 2.3 *The interaction between layers in the OSI model*

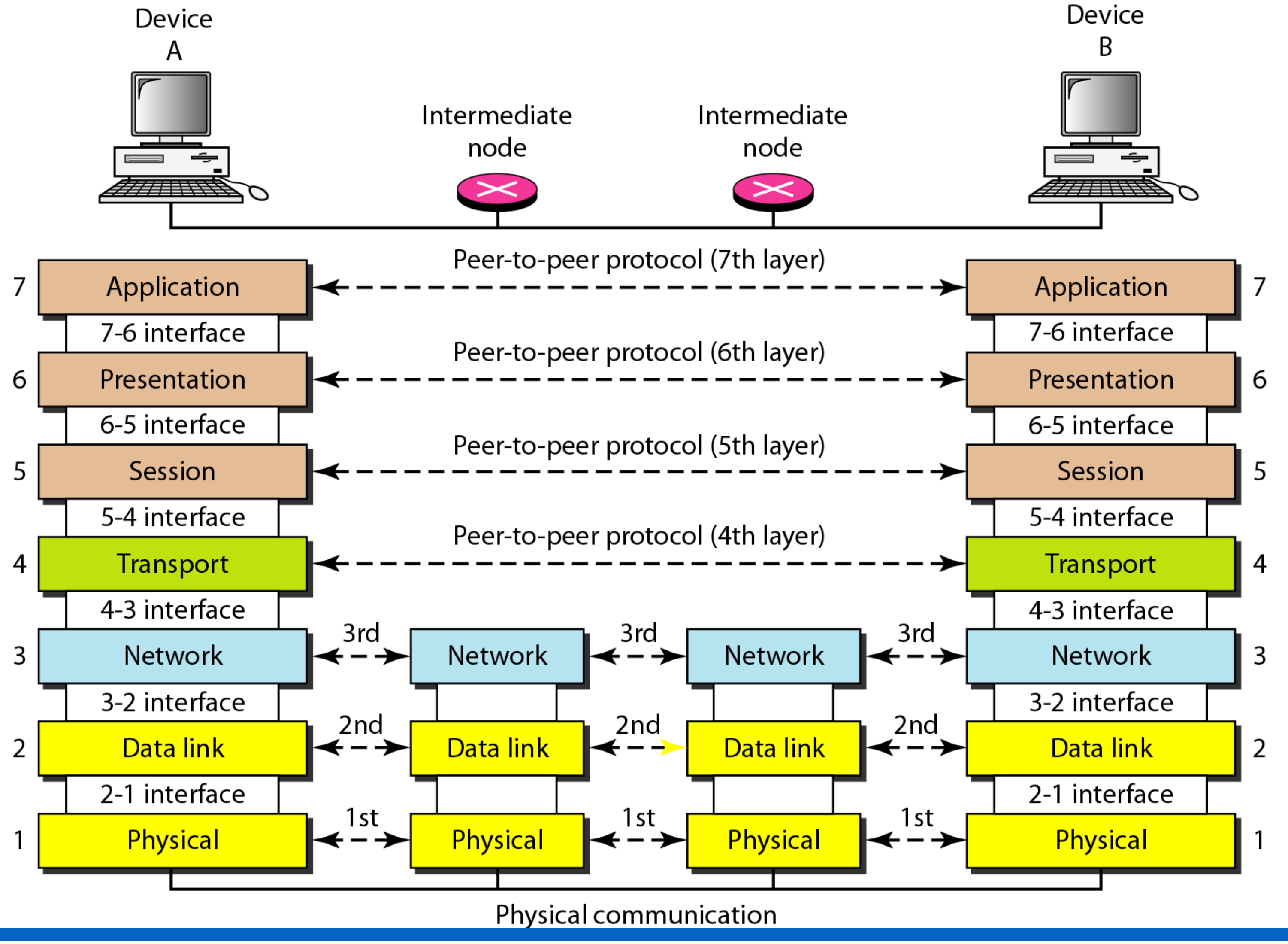
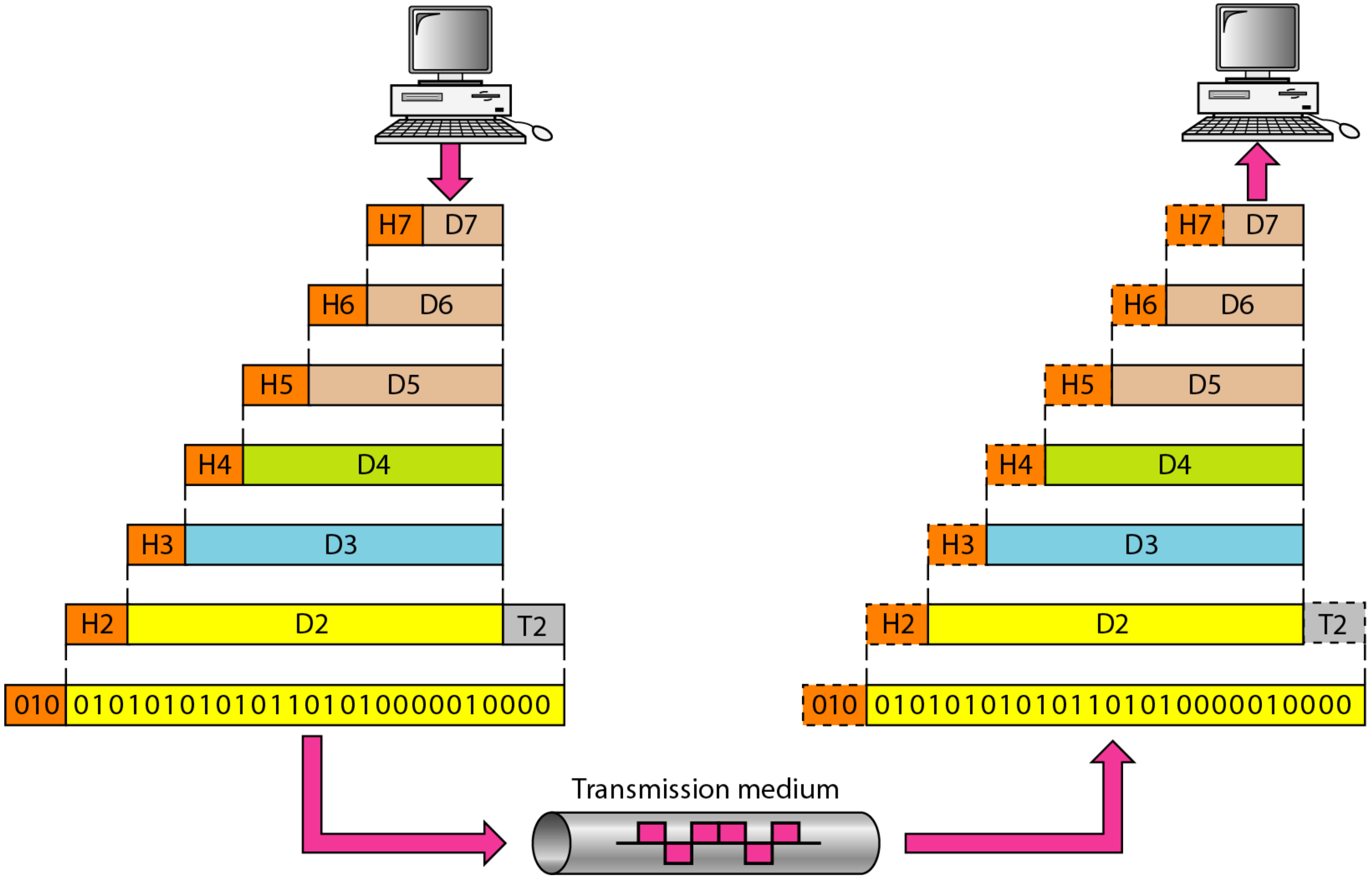


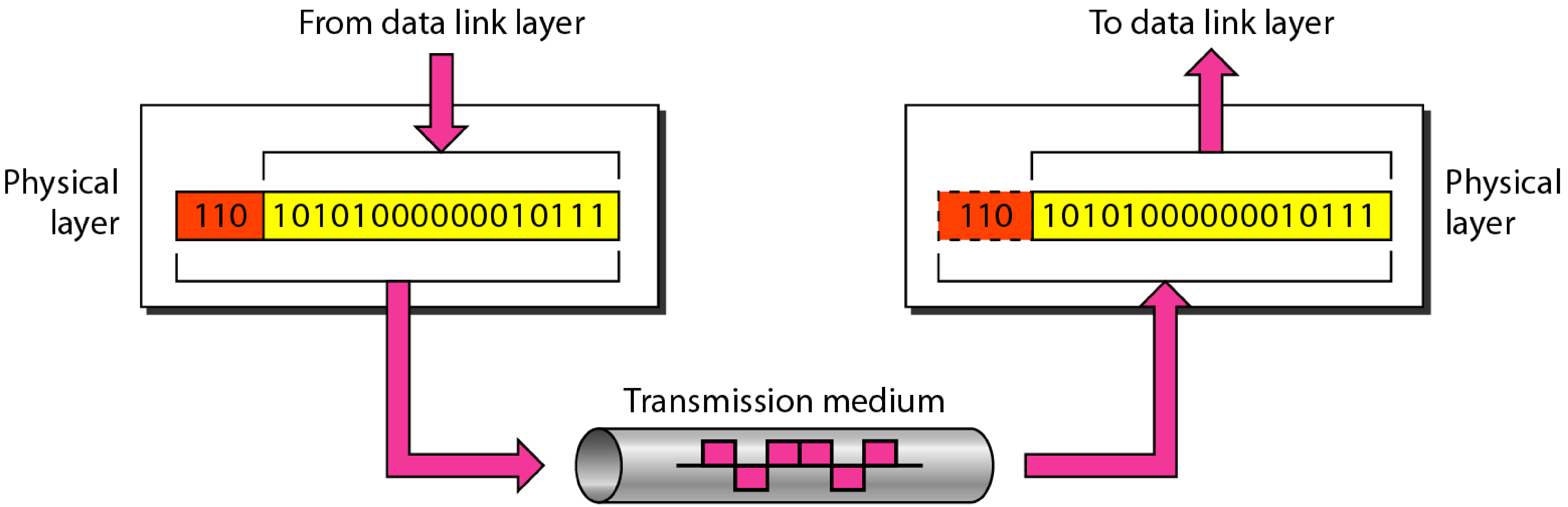
Figure 2.4 *An exchange using the OSI model*



2-3 LAYERS IN THE OSI MODEL

In this section we briefly describe the functions of each layer in the OSI model.

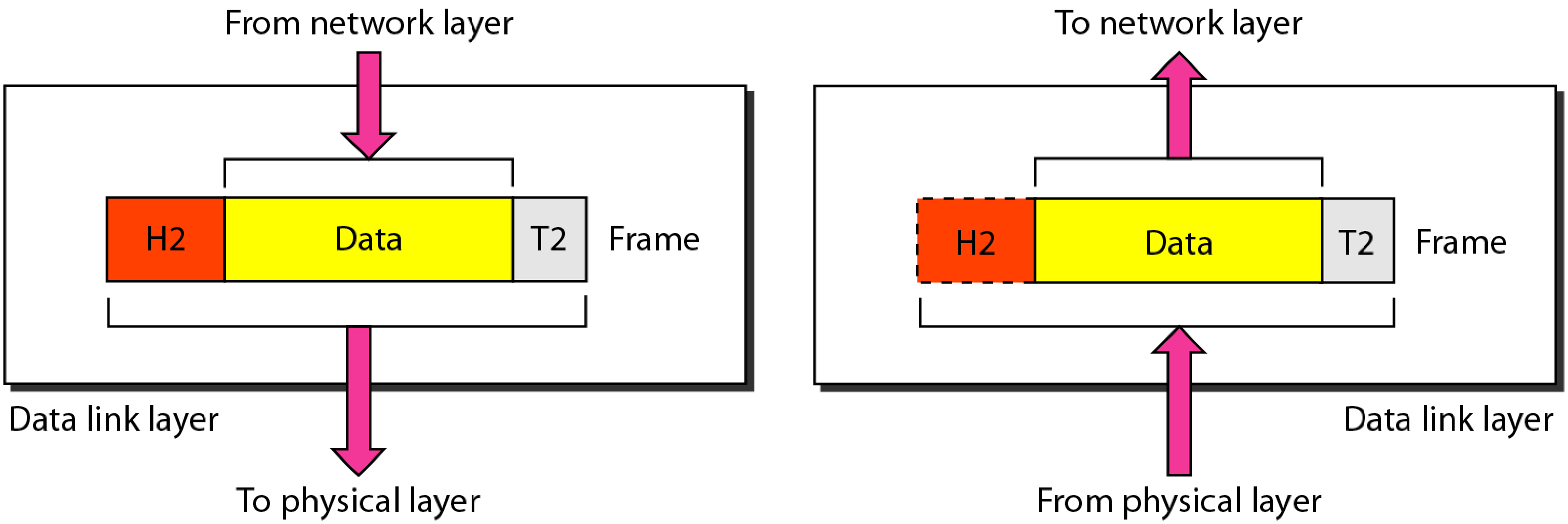
Figure 2.5 *Physical layer*



Note

The physical layer is responsible for movements of individual bits from one hop (node) to the next.

Figure 2.6 *Data link layer*



Note

The data link layer is responsible for moving frames from one hop (node) to the next.

Figure 2.7 *Hop-to-hop delivery*

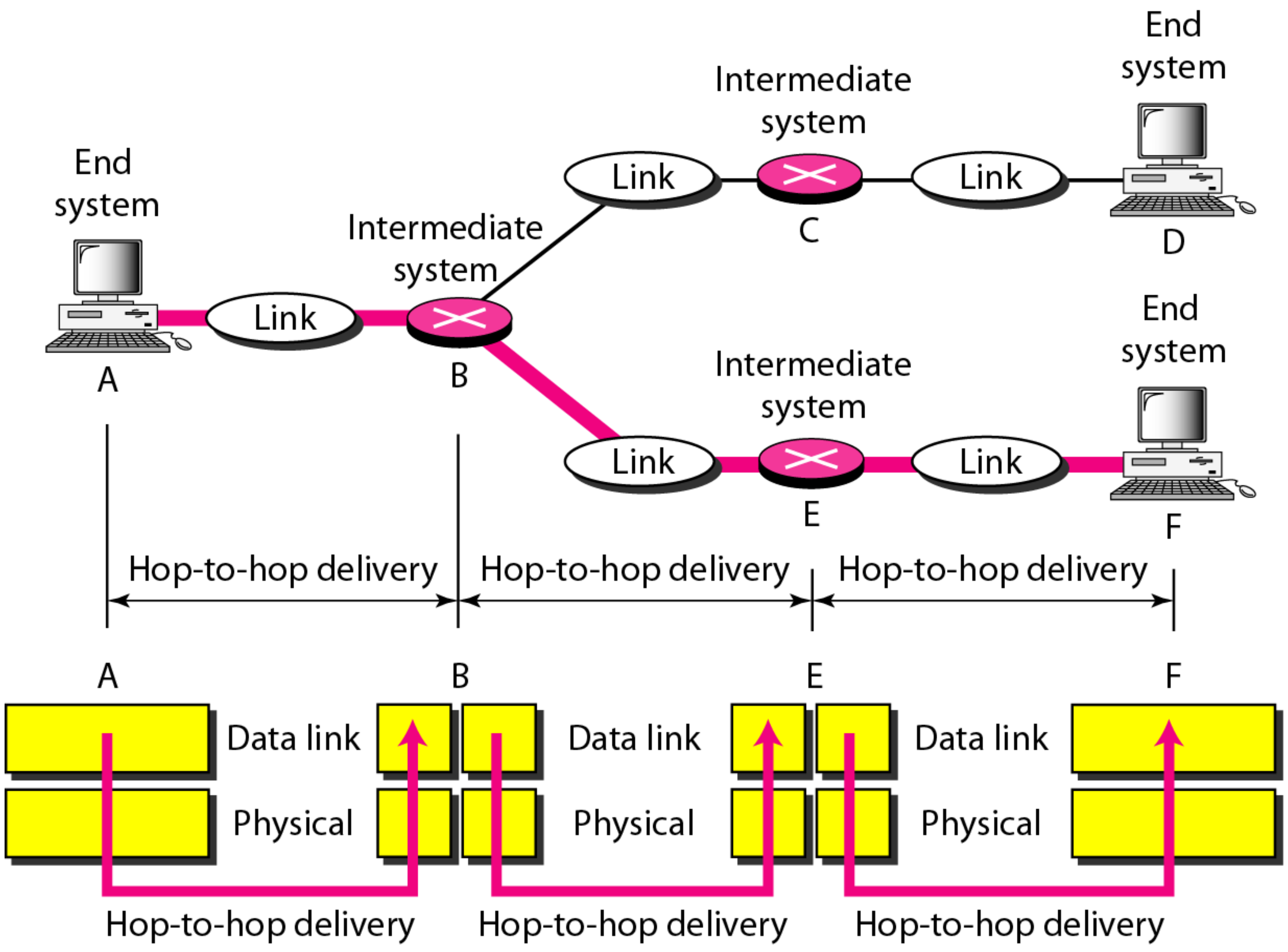
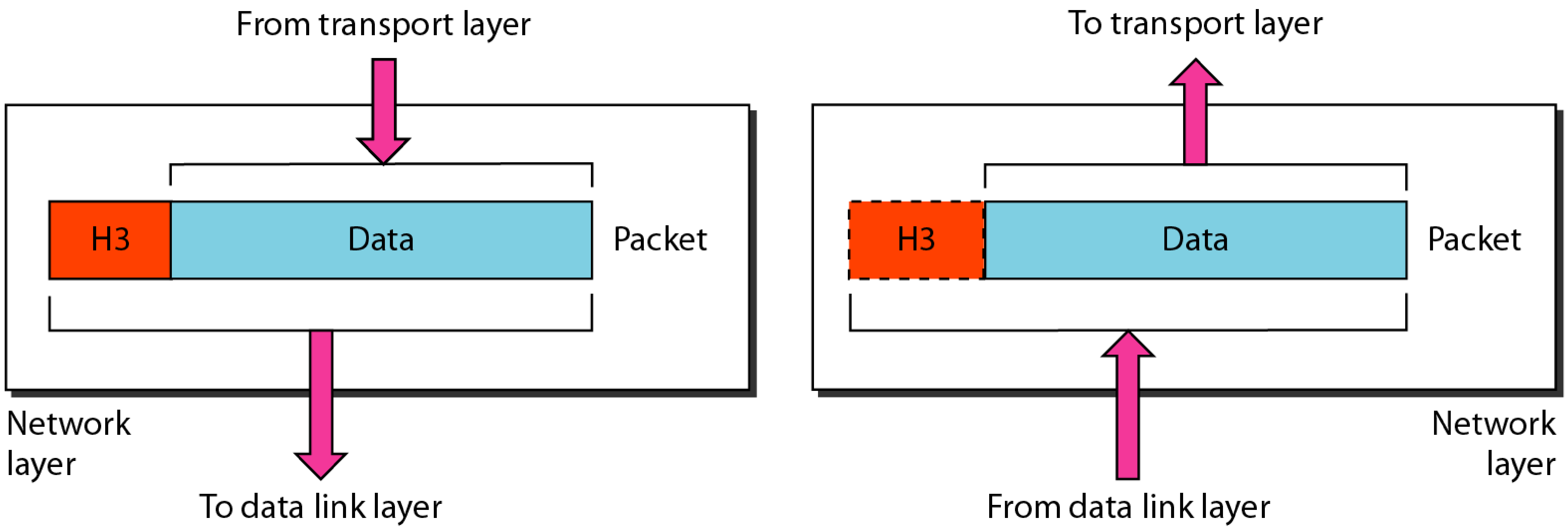


Figure 2.8 *Network layer*



Note

The network layer is responsible for the delivery of individual packets from the source host to the destination host.

Figure 2.9 *Source-to-destination delivery*

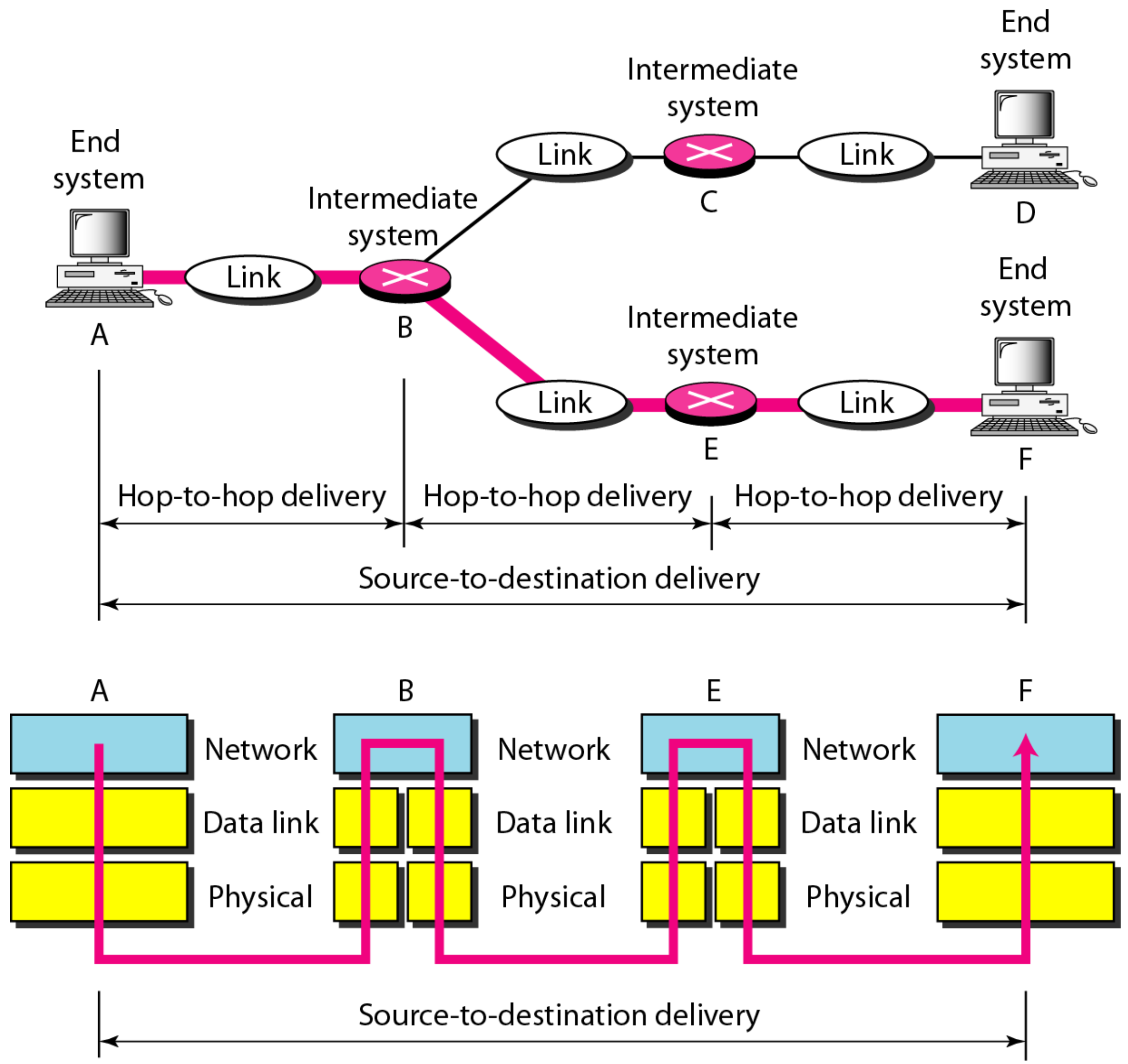
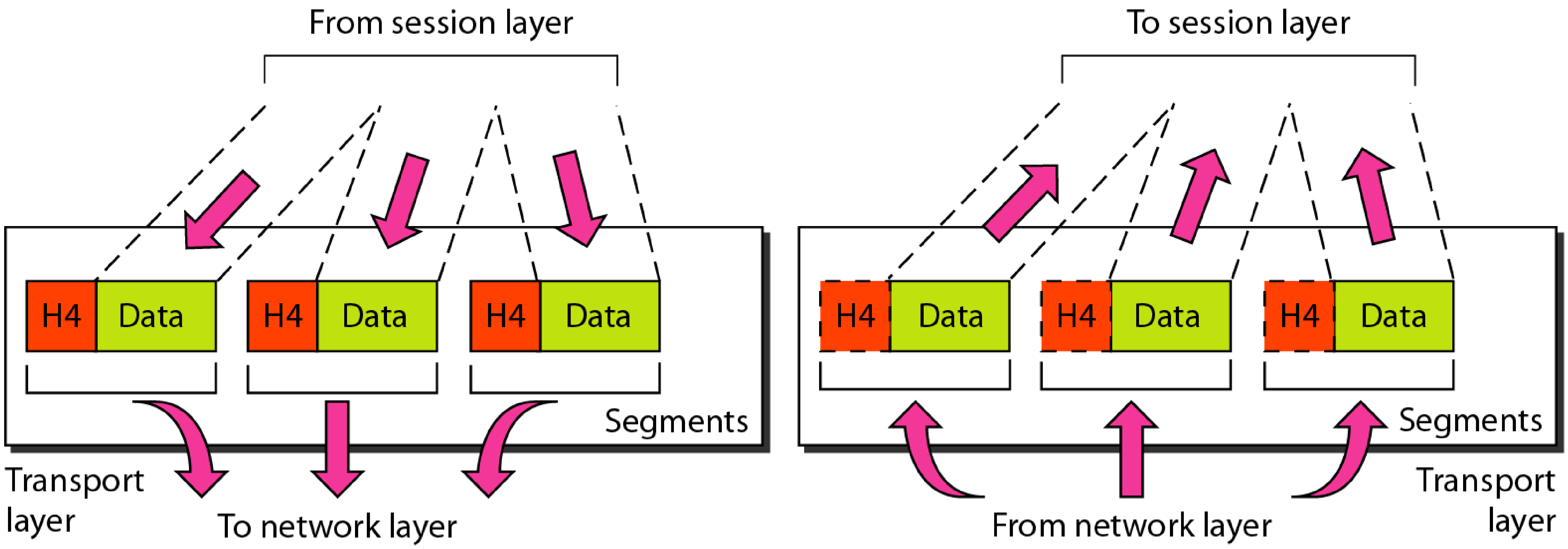


Figure 2.10 *Transport layer*



Note

The transport layer is responsible for the delivery of a message from one process to another.

Figure 2.11 *Reliable process-to-process delivery of a message*

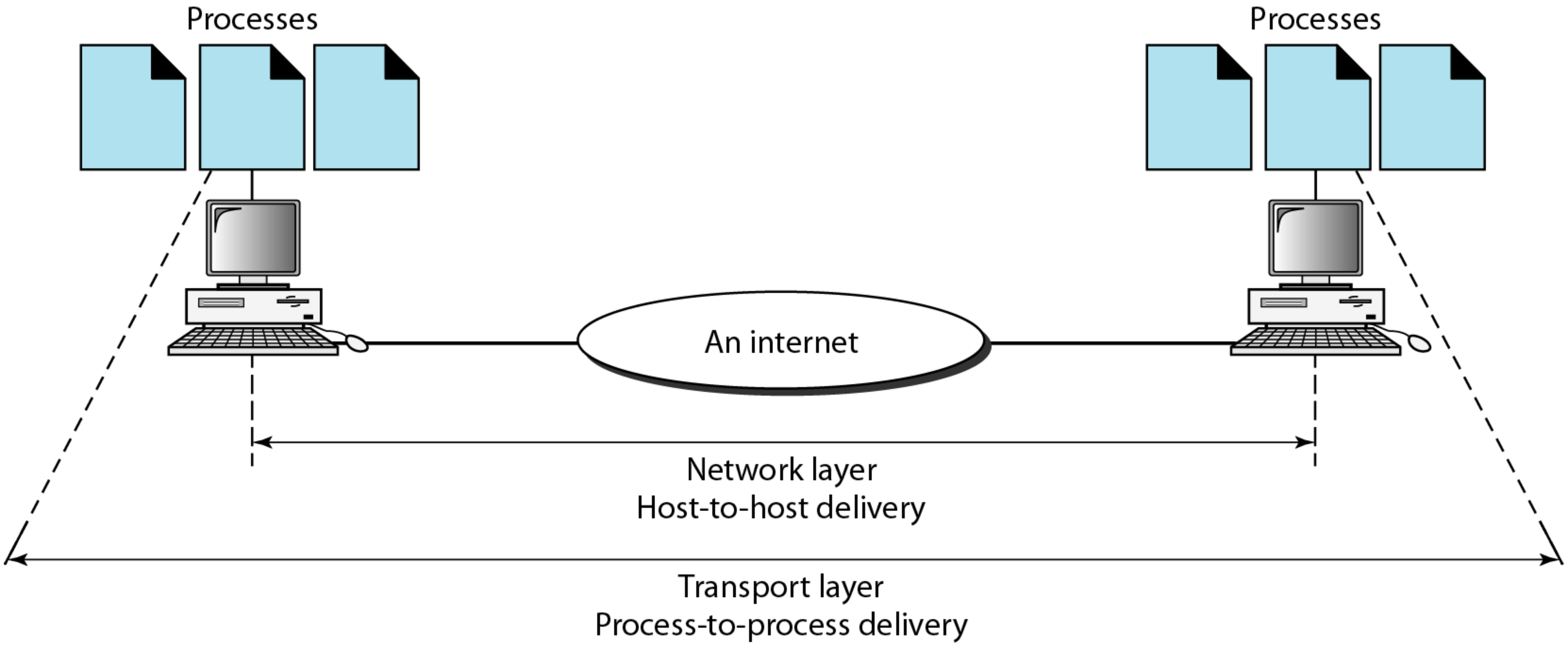
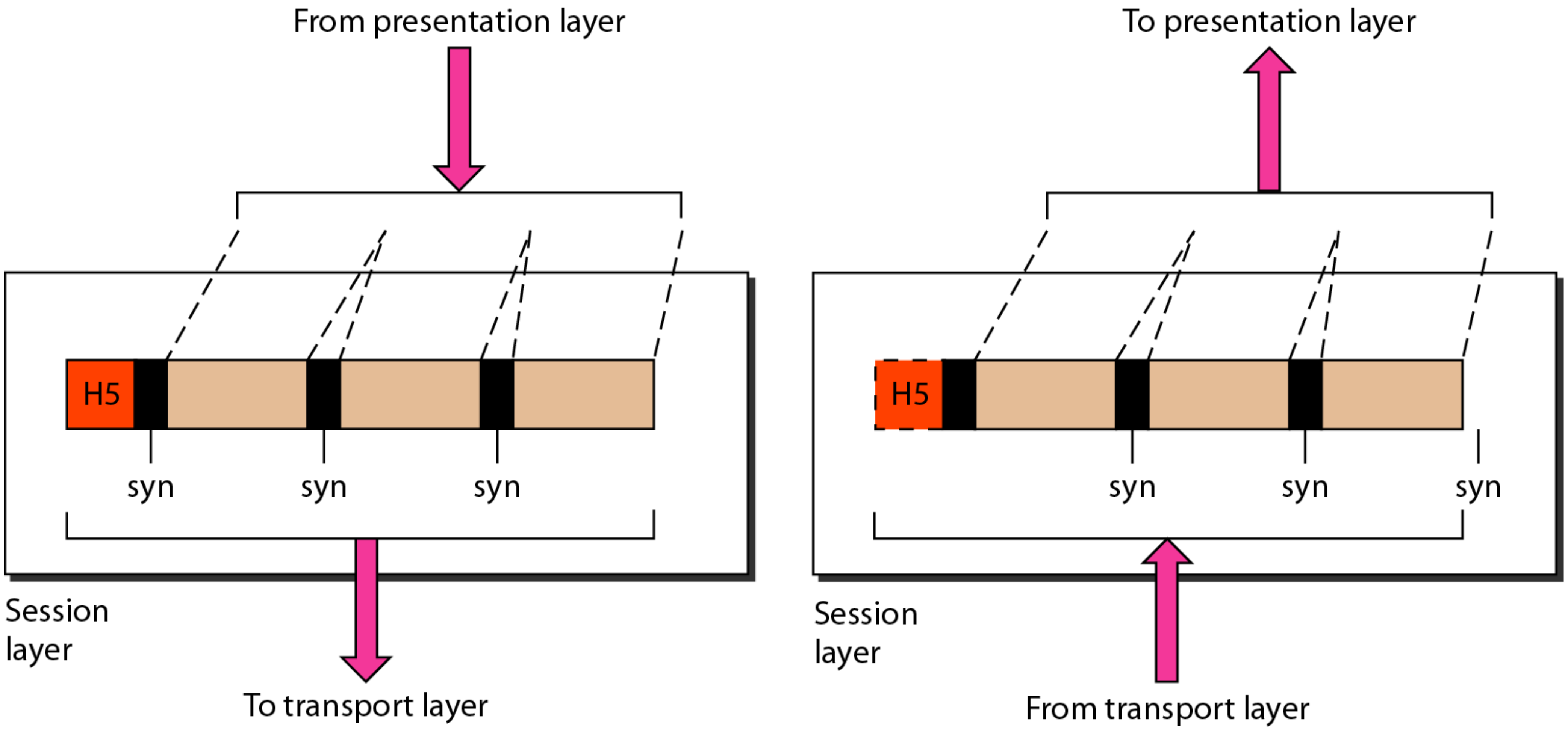


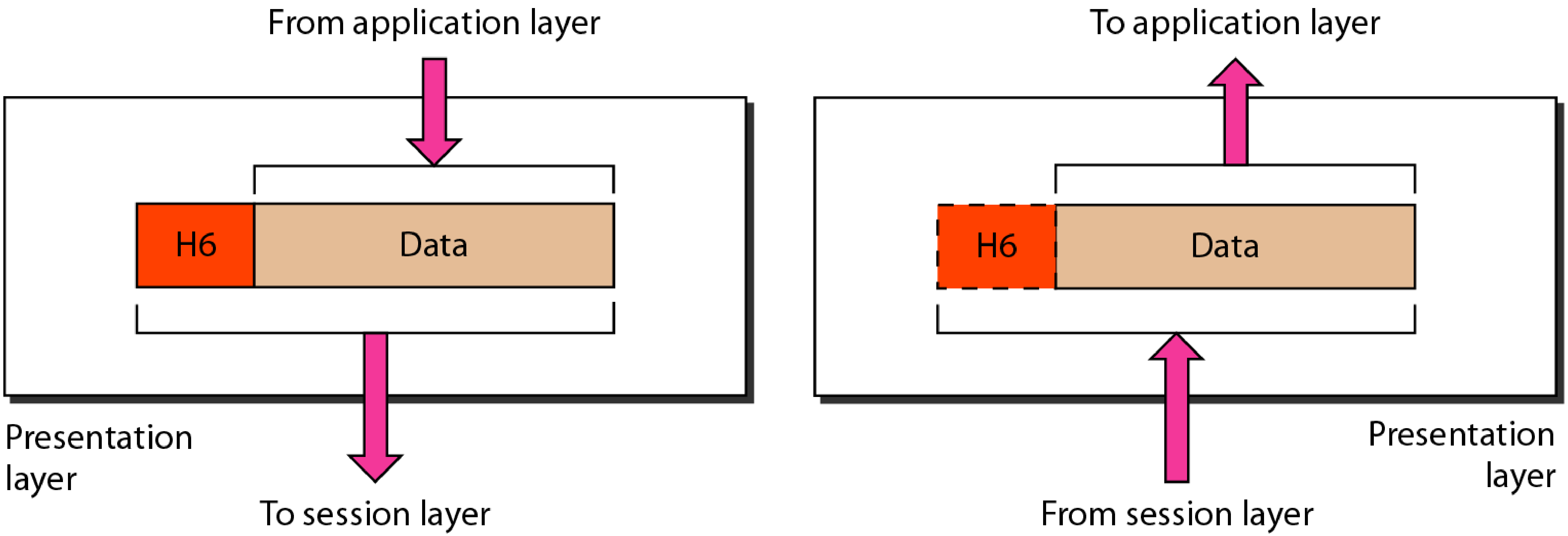
Figure 2.12 *Session layer*



Note

The session layer is responsible for dialog control and synchronization.

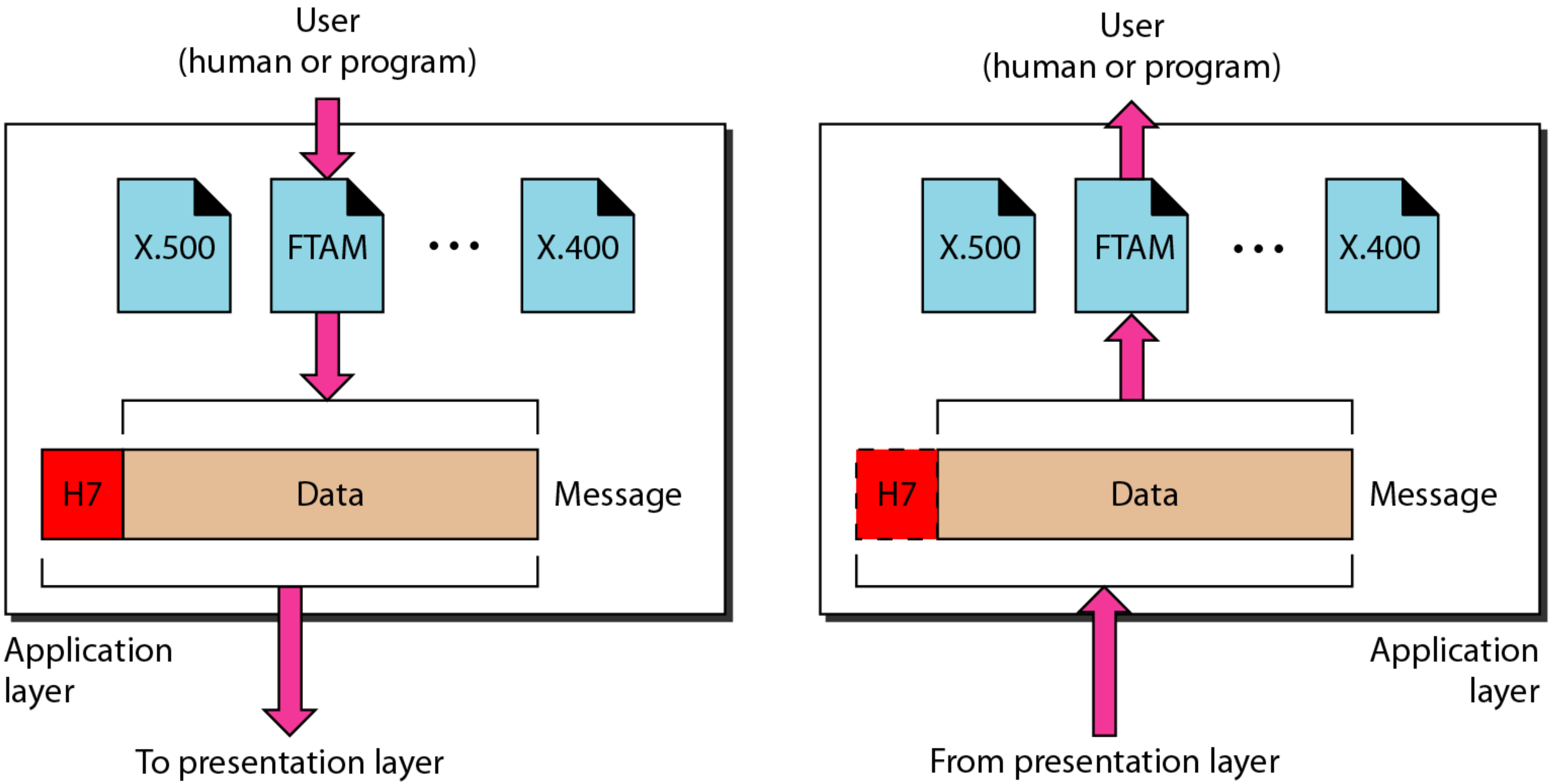
Figure 2.13 *Presentation layer*



Note

The presentation layer is responsible for translation, compression, and encryption.

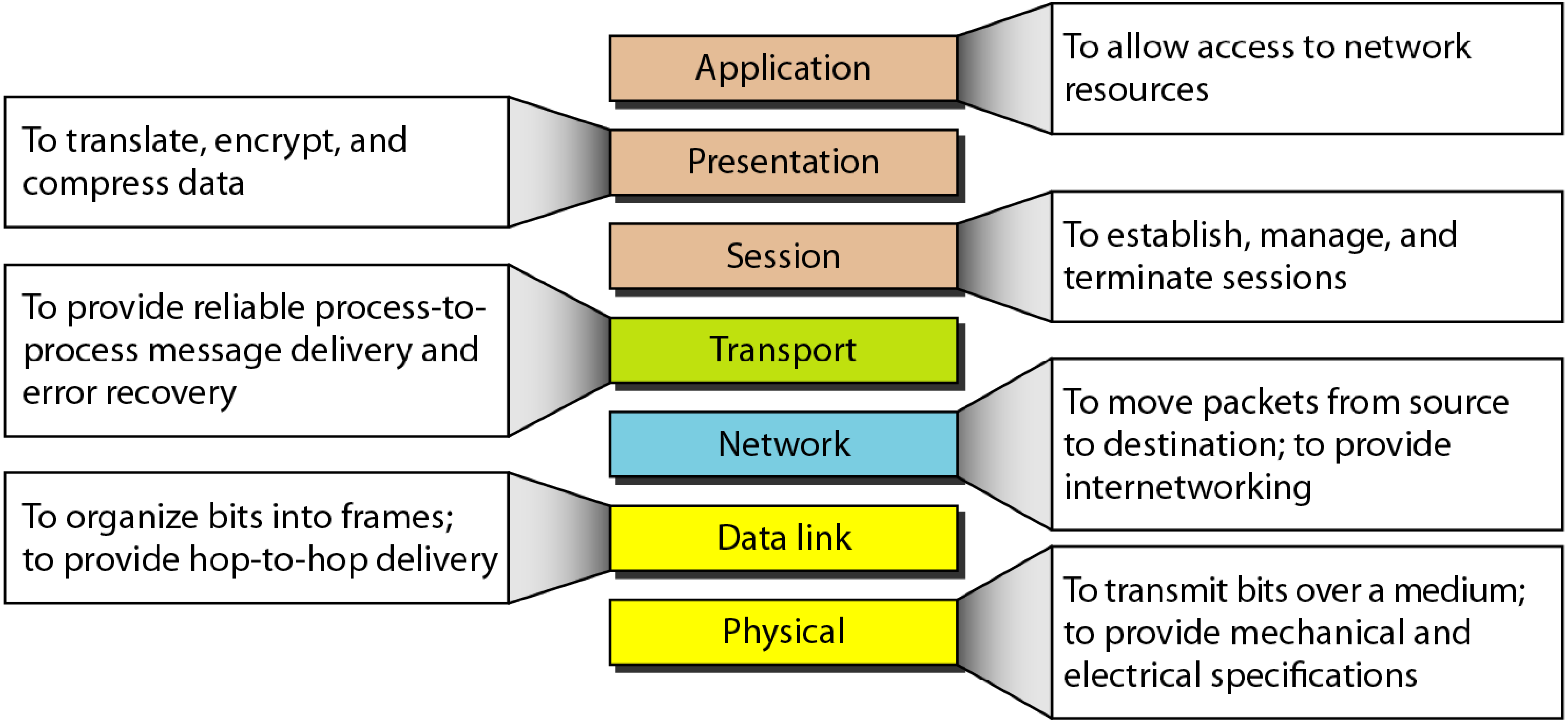
Figure 2.14 *Application layer*



Note

The application layer is responsible for providing services to the user.

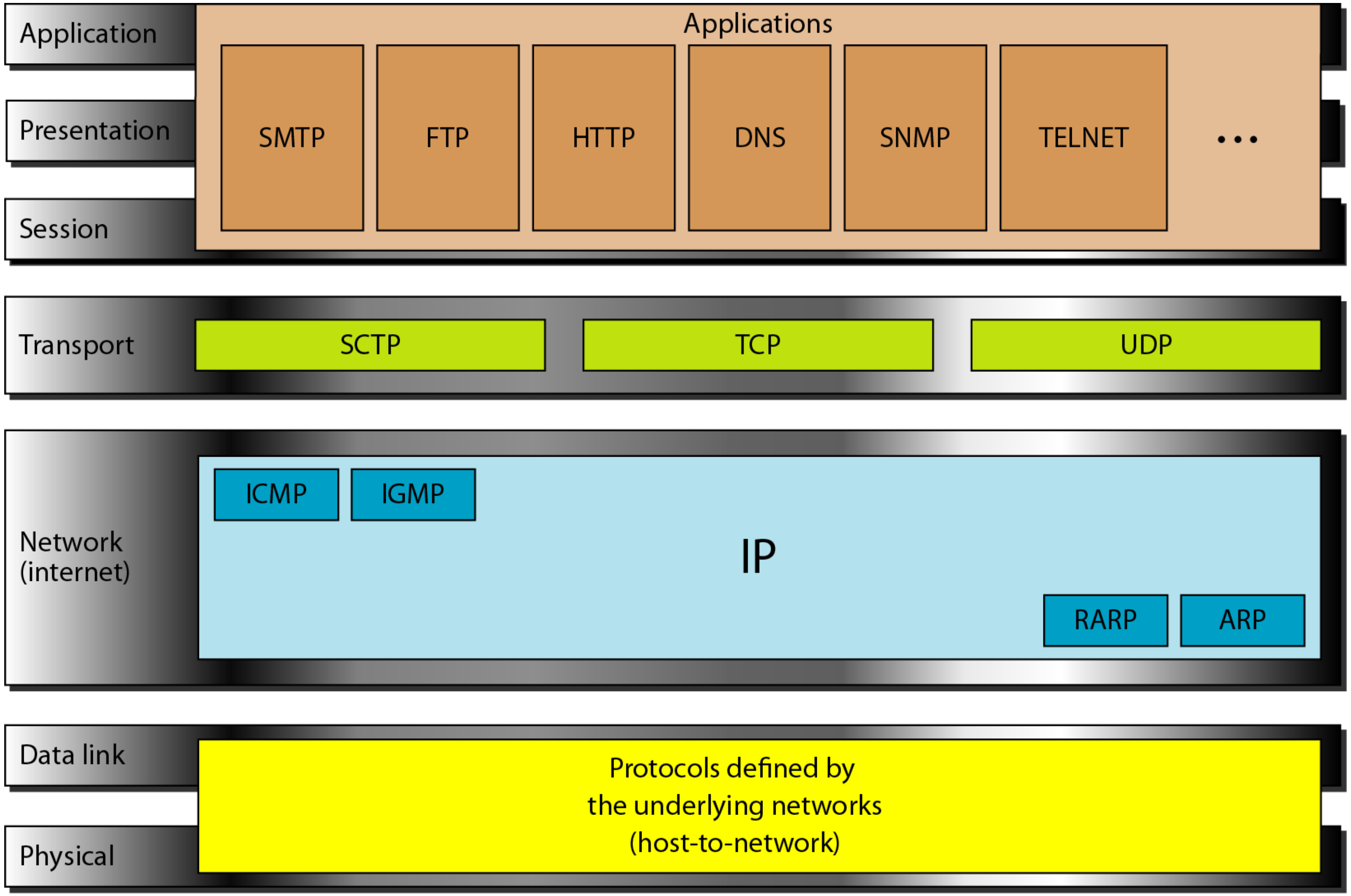
Figure 2.15 *Summary of layers*



2-4 TCP/IP PROTOCOL SUITE

- The layers in the TCP/IP protocol suite do not exactly match those in the OSI model.
- The original TCP/IP protocol suite was defined as having four layers: host-to-network, internet, transport, and application.
- However, when TCP/IP is compared to OSI, we can say that the TCP/IP protocol suite is made of five layers:
  - Physical,
  - Data link,
  - Network,
  - Transport, and
  - Application.

Figure 2.16 *TCP/IP and OSI model*



2-5 ADDRESSING

- Four levels of addresses are used in an internet employing the TCP/IP protocols:
    - Physical,
    - Logical,
    - Port, and
    - Specific.

Figure 2.17 *Addresses in TCP/IP*

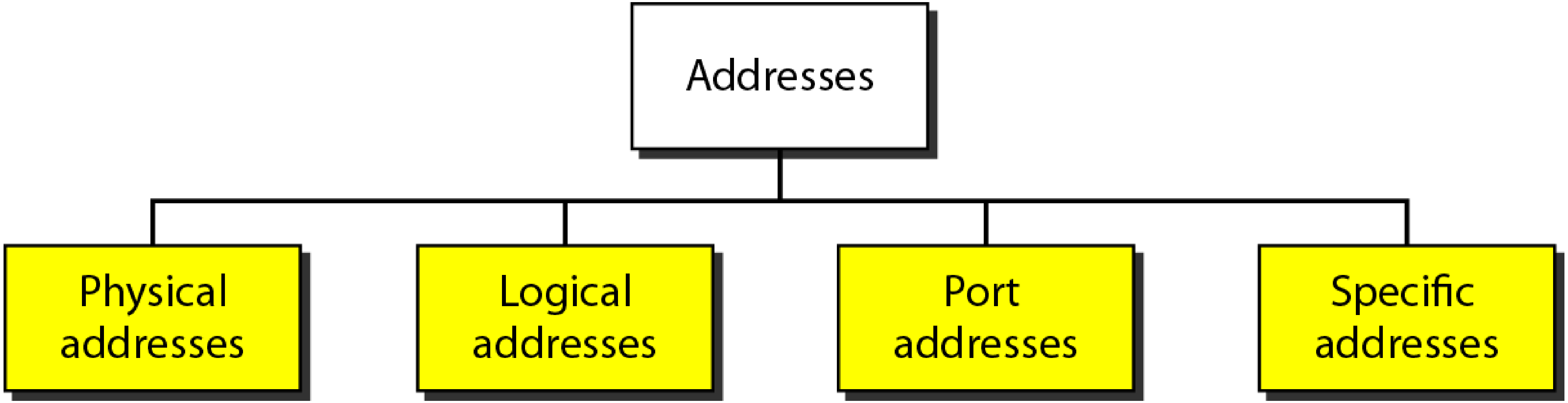
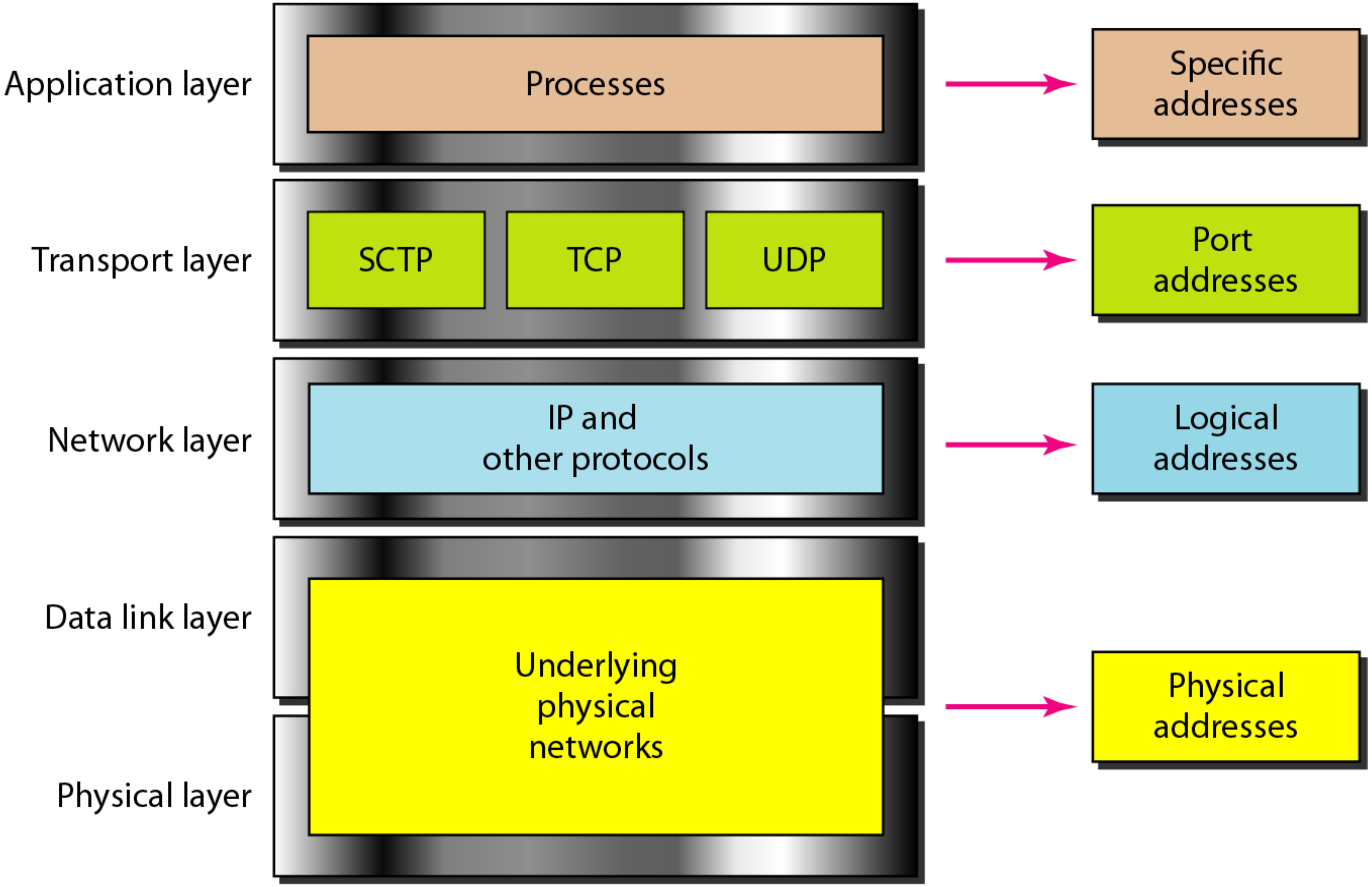


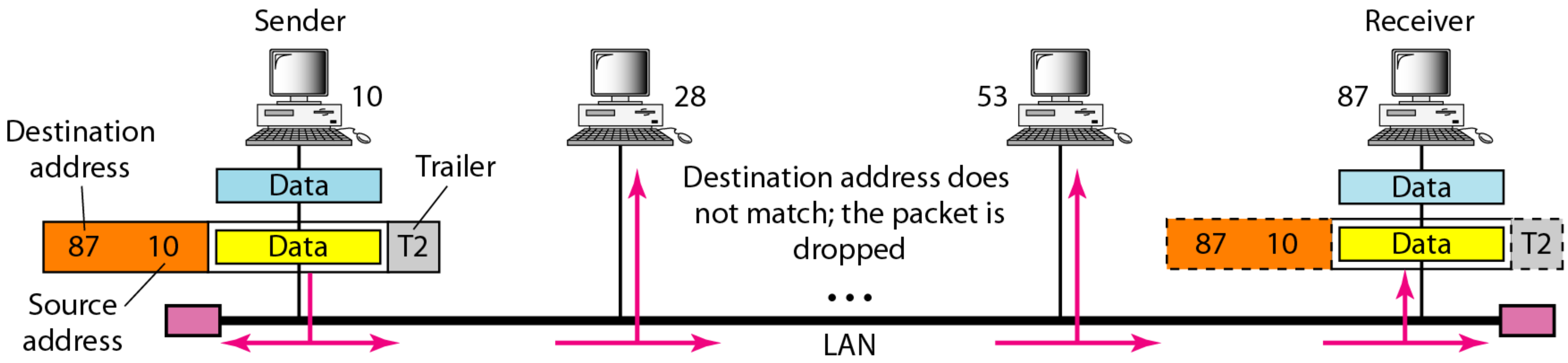
Figure 2.18 *Relationship of layers and addresses in TCP/IP*



Example 2.1

- *In Figure 2.19 a node with physical address 10 sends a frame to a node with physical address 87.*
- *The two nodes are connected by a link (bus topology LAN).*
- *As the figure shows, the computer with physical address 10 is the sender, and the computer with physical address 87 is the receiver.*

Figure 2.19 *Physical addresses*



Example 2.2

Most local-area networks use a 48-bit (6-byte) physical address written as 12 hexadecimal digits; every byte (2 hexadecimal digits) is separated by a colon, as shown below:

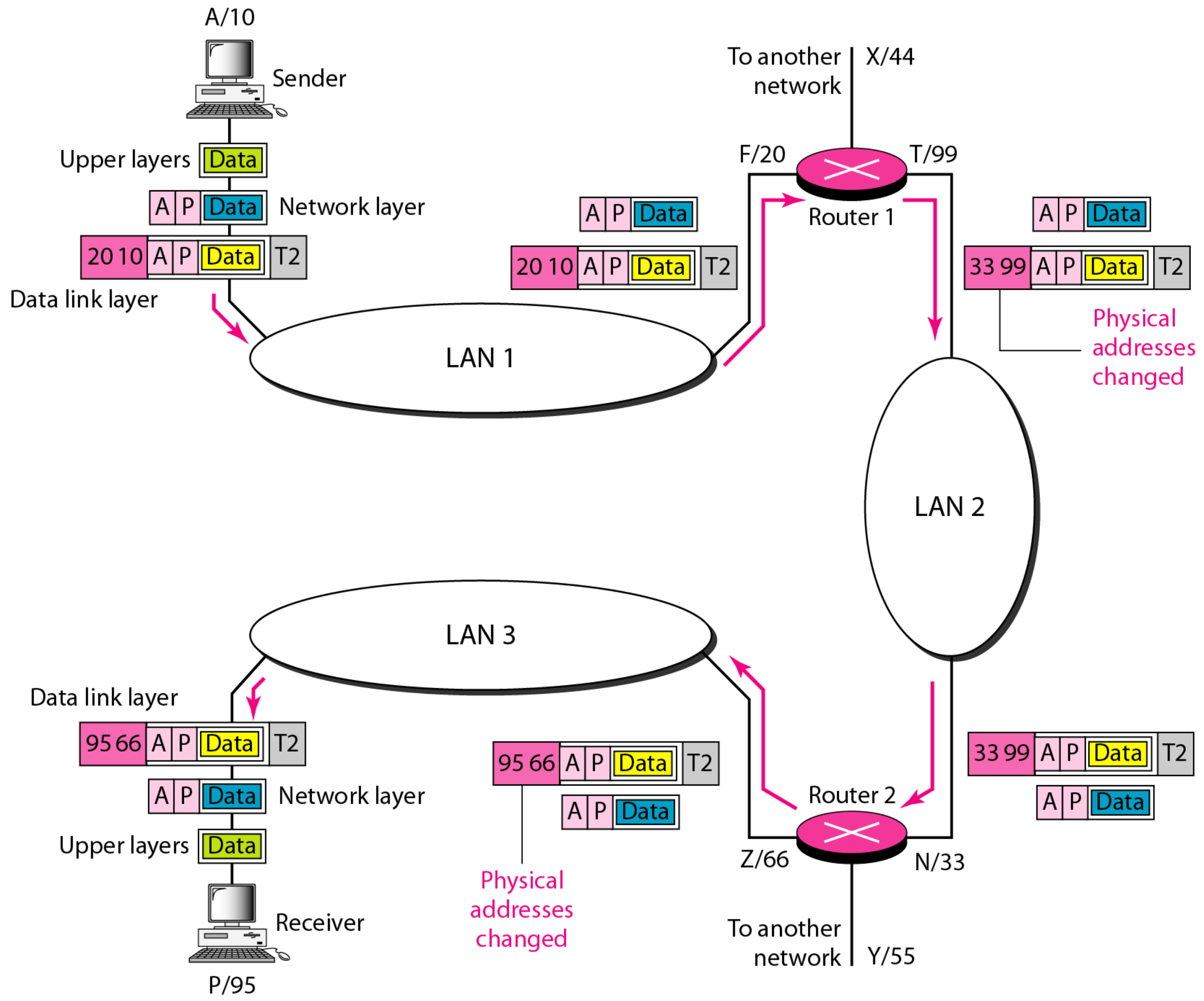
07:01:02:01:2C:4B

A 6-byte (12 hexadecimal digits) physical address.

Example 2.3

- *Figure 2.20 shows a part of an internet with two routers connecting three LANs.*
- *Each device (computer or router) has a pair of addresses (logical and physical) for each connection.*
- *In this case, each computer is connected to only one link and therefore has only one pair of addresses.*
- *Each router, however, is connected to three networks (only two are shown in the figure).*
- *So each router has three pairs of addresses, one for each connection.*

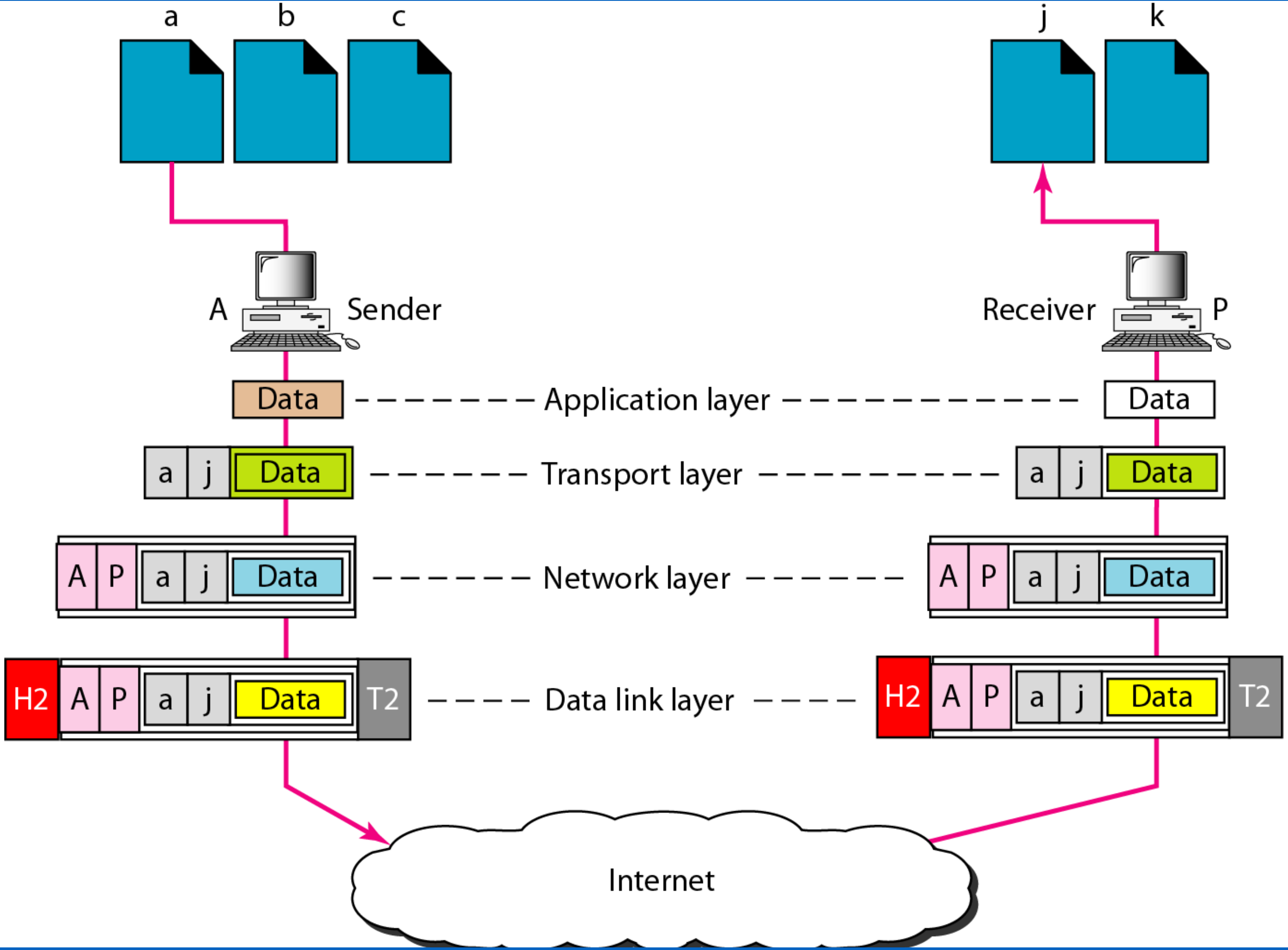
Figure 2.20 *IP addresses*



Example 2.4

- *Figure 2.21 shows two computers communicating via the Internet.*
- *The sending computer is running three processes at this time with port addresses a, b, and c.*
- *The receiving computer is running two processes at this time with port addresses j and k.*
- *Process a in the sending computer needs to communicate with process j in the receiving computer.*
- *Note that although physical addresses change from hop to hop, logical and port addresses remain the same from the source to destination.*

Figure 2.21 *Port addresses*



Note

The physical addresses will change from hop to hop,
but the logical addresses usually remain the same.

Example 2.5

A port address is a 16-bit address represented by one decimal number as shown.

753

A 16-bit port address represented as one single number.